

CHAPTER 3

DEAD LOADS, SOIL LOADS, AND HYDROSTATIC PRESSURE

3.1 DEAD LOADS

3.1.1 Definition. Dead loads consist of the weight of all materials of construction incorporated into the building including, but not limited to, walls, floors, roofs, ceilings, stairways, built-in partitions, finishes, cladding, and other similarly incorporated architectural and structural items and fixed service equipment, including the weight of cranes and material handling systems.

3.1.2 Weights of Materials and Constructions. In determining dead loads for purposes of design, the actual weights of materials and constructions shall be used, provided that in the absence of definite information, values approved by the Authority Having Jurisdiction shall be used.

3.1.3 Weight of Fixed Service Equipment. In determining dead loads for purposes of design, the weight of fixed service equipment, including the maximum weight of the contents of fixed service equipment, shall be included. The components of fixed service equipment that are variable, such as liquid contents and movable trays, shall not be used to counteract forces causing

overturning, sliding, and uplift conditions in accordance with Section 1.3.6.

EXCEPTIONS:

1. Where force effects are the result of the presence of the variable components, the components are permitted to be used to counter those load effects. In such cases, the structure shall be designed for force effects with the variable components present and with them absent.
2. For the calculation of seismic force effects, the components of fixed service equipment that are variable, such as liquid contents and movable trays, need not exceed those expected during normal operation.

3.1.4 Vegetative and Landscaped Roofs. The weight of all landscaping and hardscaping materials shall be considered as dead load. The weight shall be computed considering both fully saturated soil and drainage layer materials and fully dry soil and drainage layer materials to determine the most severe load effects on the structure.

Table 3.2-1 Design Lateral Soil Load

Description of Backfill Material	Unified Soil Classification	Design Lateral Soil Load ^a psf per foot of depth (kN/m ² per meter of depth)
Well-graded, clean gravels, gravel-sand mixes	GW	35 (5.50) ^b
Poorly graded, clean gravels, gravel-sand mixes	GP	35 (5.50) ^b
Silty gravels, poorly graded gravel-sand mixes	GM	35 (5.50) ^b
Clayey gravels, poorly graded gravel-and-clay mixes	GC	45 (7.07) ^b
Well-graded, clean sands; gravel-sand mixes	SW	35 (5.50) ^b
Poorly graded, clean sands, sand-gravel mixes	SP	35 (5.50) ^b
Silty sands, poorly graded sand-silt mixes	SM	45 (7.07) ^b
Sand-silt clay mix with plastic fines	SM-SC	85 (13.35) ^c
Clayey sands, poorly graded sand-clay mixes	SC	85 (13.35) ^c
Inorganic silts and clayey silts	ML	85 (13.35) ^c
Mixture of inorganic silt and clay	ML-CL	85 (13.35) ^c
Inorganic clays of low to medium plasticity	CL	100 (15.71) ^d
Organic silts and silt-clays, low plasticity	OL	^d
Inorganic clayey silts, elastic silts	MH	^d
Inorganic clays of high plasticity	CH	^d
Organic clays and silty clays	OH	^d

^aDesign lateral soil loads are given for moist conditions for the specified soils at their optimum densities. Actual field conditions shall govern. Submerged or saturated soil pressures shall include the weight of the buoyant soil plus the hydrostatic loads.

^bFor relatively rigid walls, as when braced by floors, the design lateral soil load shall be increased for sand and gravel type soils to 60 psf (9.43 kN/m²) per foot (meter) of depth. Basement walls extending not more than 8 ft (2.44 m) below grade and supporting light floor systems are not considered as being relatively rigid walls.

^cFor relatively rigid walls, as when braced by floors, the design lateral load shall be increased for silt and clay type soils to 100 psf (15.71 kN/m²) per foot (meter) of depth. Basement walls extending not more than 8 ft (2.44 m) below grade and supporting light floor systems are not considered as being relatively rigid walls.

^dUnsuitable as backfill material.

3.1.5 Solar Panels. The weight of solar panels, their support system, and ballast shall be considered as dead load.

3.2 SOIL LOADS AND HYDROSTATIC PRESSURE

3.2.1 Lateral Pressures. Structures below grade shall be designed to resist lateral soil loads from adjacent soil. If lateral soil loads are not given in a geotechnical report approved by the Authority Having Jurisdiction, then the lateral soil loads specified in Table 3.2-1 shall be used as the minimum design lateral soil loads. Where applicable, lateral pressure from fixed or moving surcharge loads shall be added to the lateral soil loads. When a portion or the whole of the adjacent soil is below a free-water surface, computations shall be based upon the weight of the soil diminished by buoyancy, plus full hydrostatic pressure.

The lateral pressure shall be increased if expansive soils are present at the site, as determined by a geotechnical investigation.

3.2.2 Uplift Loads on Floors and Foundations. Basement floors, slabs on ground, foundations, and similar approximately horizontal elements below grade shall be designed to resist uplift loads where applicable. The upward pressure of water shall be taken as the full hydrostatic pressure applied over the entire area. The hydrostatic load shall be measured from the underside of the construction.

Foundations, slabs on ground, and other components placed on expansive soils shall be designed to tolerate the movement or resist the upward loads caused by the expansive soils, or the expansive soil shall be removed or stabilized around and beneath the structure.

3.3 CONSENSUS STANDARDS AND OTHER REFERENCED DOCUMENTS

No consensus standards and other documents that shall be considered part of this standard are referenced in this chapter.